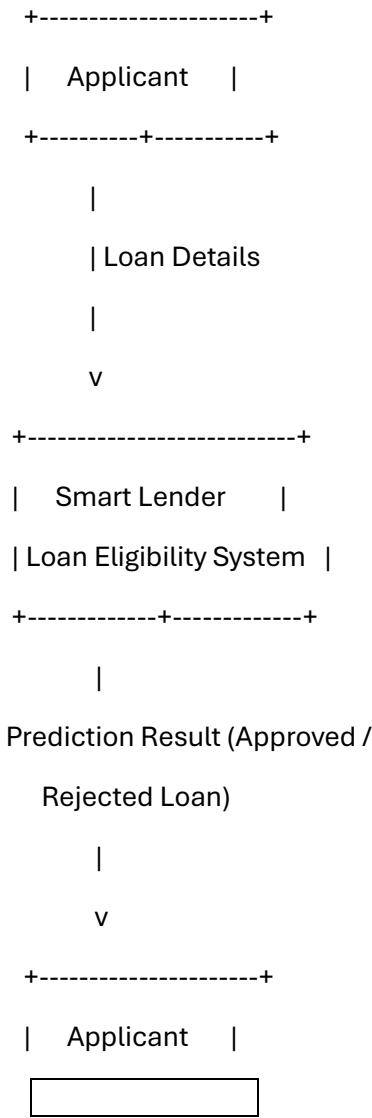


Data Flow Diagram

Introduction

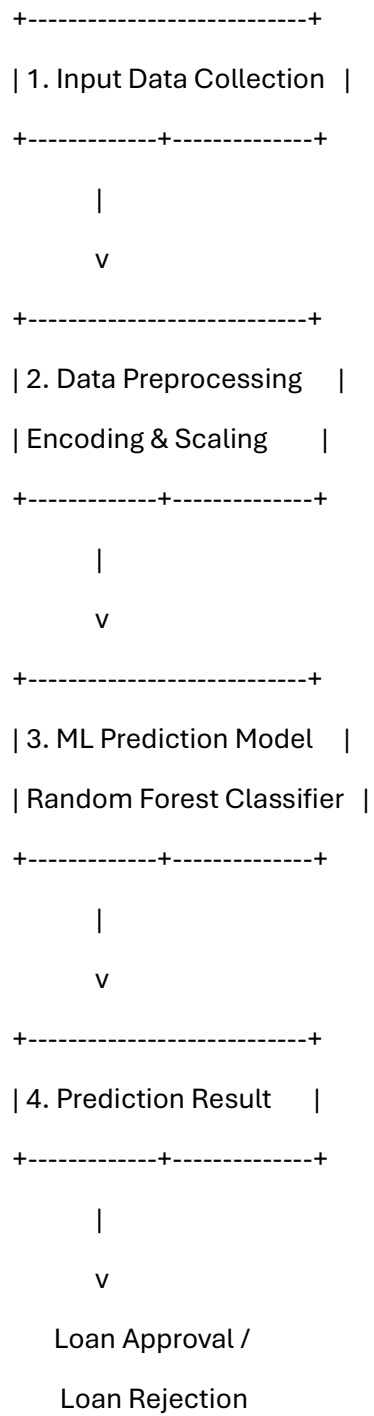
A Data Flow Diagram (DFD) illustrates how data moves through the Smart Lender system. It shows the interaction between users, the web application, the machine learning model, and the prediction result

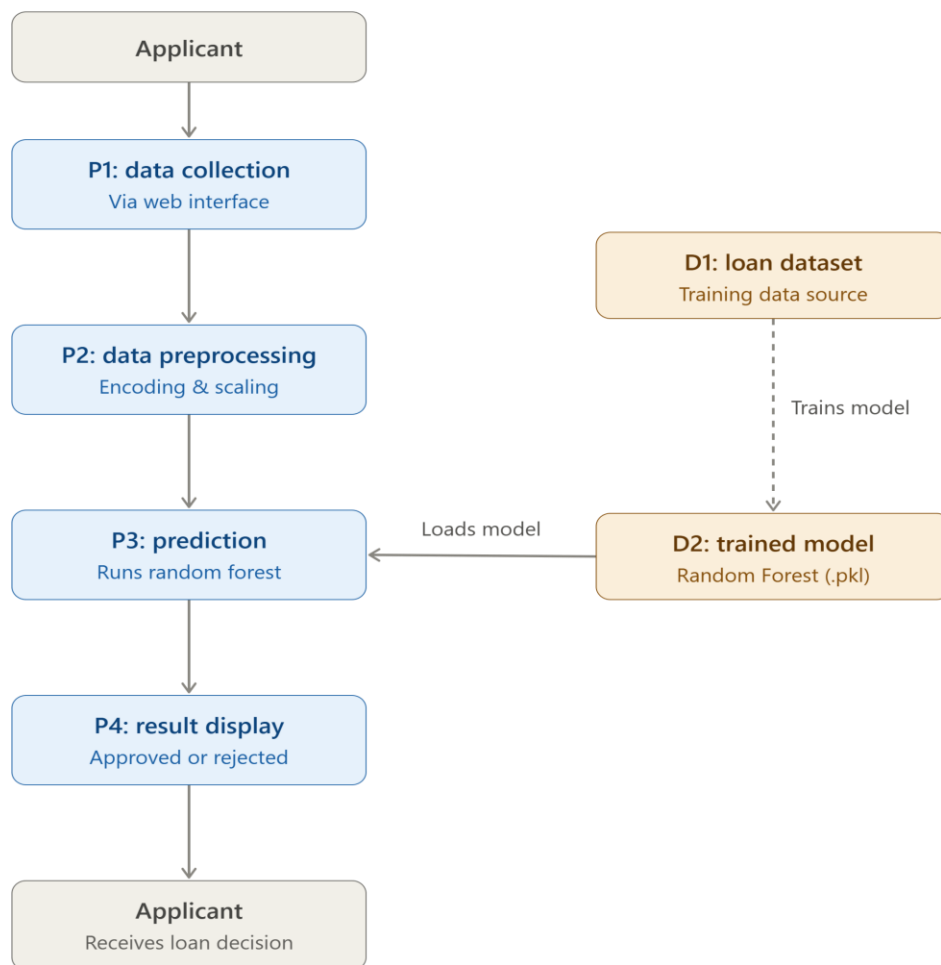
Level 0 DFD (Context Diagram)



Level 1 DFD







Data Stores

D1 – Loan Dataset

Contains historical applicant information used to train the machine learning model.

D2 – Trained Machine Learning Model

Stores the serialized Random Forest model (saved_model.pkl) used for prediction.

External Entity

Applicant

Provides loan application details and receives the loan eligibility prediction.

Processes

Process 1 – Data Collection

Receives applicant information through the web interface.

Process 2 – Data Preprocessing

Performs encoding, scaling, and formatting of the input data.

Process 3 – Prediction

Uses the trained Random Forest model to predict loan eligibility.

Process 4 – Result Display

Displays the prediction result (Approved or Rejected) on the web application.

Conclusion

The Data Flow Diagram demonstrates how applicant information is collected, processed, analyzed using the machine learning model, and returned as an instant loan eligibility prediction.